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Thick fingers?

Projected Capacitive technology can detect touches made on a screen up to 2 cm thick

The Touchables

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Touchscreens (TS) have been around for a long time, and quite inconspicuously so, till the iPhone amazed people with the multi-touch capabilities of its interface. Now, if you are dismissively wondering *how long before?* you will be surprised to hear *more than a quarter of a century*. Before we proceed, we need to clearly understand what qualifies as a TS. A largely satisfactory definition would be – any transparent interface that generates an input either on touch or proximity with another object can be termed a TS. Now, those of you who have encountered a TS before, mostly at kiosks at the ATM or railway station, will recall that during

some encounters, mere touch was not sufficient and significant pressure had to be employed to get a response. This would disqualify that interface as a TS in light of our definition.

Technologies

Broadly there are two parts to the whole touch experience. First, and most conspicuously, the mechanical interface itself - that merely generates the input; and second, the underlying software - that interprets the input into something meaningful. It is important to note, that the software plays an important role and it can make and mar the touch experience irrespective of the quality of the mechanical components. For example, a slide of a finger on the interface is interpreted as a “flick” by the software.

Resistive TS

This is the most mature among the technologies and also the cheapest and the most widely used. Here, the TS is constructed using two flexible films whose proximal surfaces are coated with Indium Tin Oxide (ITO). ITO is a solution of Indium oxide and Tin oxide, which is conductive and, in thin layers, transparent. The two films are microscopically separated by an array of insulating spacers. In working, when the upper flexible layer is pressed, it touches the lower layer which completes a circuit and the location of the point is computed.

Not so fast! The technique with which the point of touch is computed distinguishes the subtypes of this technology. So a 4 wire Resistive interface uses two wires on opposite edges of each on the top

Illustration: Shrikishna Patkar

